SOONHYUN KWON (권순현)

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RESEARCH INTERESTS

Robot manipulation and autonomy, especially Vision-Language-Action (VLA) models for robot arms and humanoid robots. At Beyond Robotics I build vision (RF-DETR) and Behavior Tree control pipelines for robot arms in ROS 2; next, I want to study replacing these hand-designed behaviors with learned VLA policies. I bring 4+ years of validation experience (HIL circuit-fault regression testing at John Deere, CAN-log analysis at Hyundai) to testing robot systems systematically before they run on real hardware.

EDUCATION

B.S. in Electrical Engineering — *Flint, MI*
Kettering University — **2017 – 2022**

- Kettering Provost Scholarship; Dean's List

EXPERIENCE

Software Engineer — *Seoul, KR*
Beyond Robotics — **Jul 2026 – Present**

- Modified Behavior Tree–based control for a dual-arm (UFactory) strawberry harvesting robot, demonstrated at the 2026 FFK exhibition.
- Independently set up the camera and built the Behavior Tree for a robot-arm pick-and-place cell.
- Integrated an RF-DETR vision model into the robot's ROS 2 perception pipeline.
- Built a ROS 2/CAN robot control GUI using rosbridge.

Software Test Engineer — *Waterloo, IA*
John Deere — **May 2023 – Oct 2025**

- Performed regression testing of embedded machine software on hardware-in-the-loop (HIL) benches using ETAS LABCAR and MATLAB, including circuit-fault testing.
- Wrote and executed test plans to verify the functional correctness of embedded software.
- Validated closed-loop features and sensor-to-actuator behavior, both automated and manual, using Vector CANalyzer, CANape, and Hardware Config.
- Ran bench tests with multidisciplinary teams to find and resolve system integration and communication issues.

Telematics Product Validation Engineer — *Superior Charter Township, MI*
Hyundai America Technical Center — **Oct 2022 – Apr 2023**

- Collected and analyzed CAN logs and app signal data to validate the Bluelink connected-car app across Hyundai, Kia, and Genesis vehicles.
- Ran vehicle-level CAN simulations and diagnosed communication and synchronization issues between vehicle modules and servers.
- Managed defect reporting in Jira and built a shared Jira repository for the validation team.

Electronics Technician Intern — *Auburn Hills, MI*
Continental — **Apr 2022 – Jul 2022**

- Verified Body Control Module (BCM) behavior and data parameters for a demo vehicle project.
- Built a custom load-box test rig that simulates vehicle loads to test smart-key functions (keyless entry and push-button start).
- Designed worst-case stress tests from schematics and logic configurations.

Software Test Engineer Co-op — *Northville, MI*
Gentherm — **Oct 2017 – Jul 2019**

- Validated closed-loop motor feedback (power seats) and thermal control systems using CANalyzer.
- Translated customer needs into technical requirements in IBM DOORS.
- Supported on-site manufacturing validation at Lear Corporation.

SKILLS

Robotics	ROS 2, Behavior Trees, rosbridge
ML / Vision	RF-DETR, PyTorch
Programming	Python, C++, MATLAB/Simulink
Tools	CAN, Vector CANalyzer/CANape, ETAS LABCAR, Jira, Rally, IBM DOORS